Inequalities on Graphs

Exam code: 9MA0

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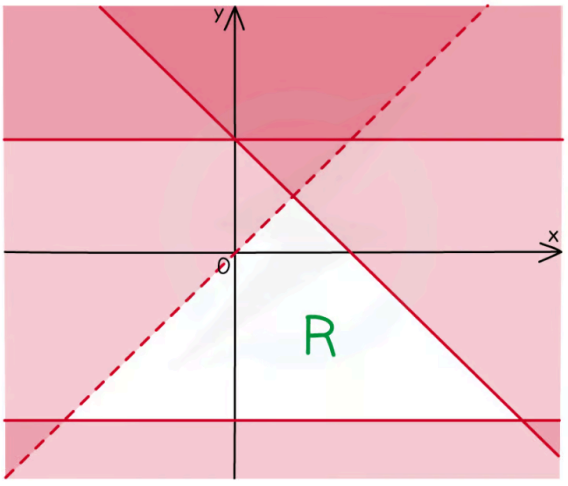
Help me understand

Inequalities on graphs

What are inequalities on graphs?

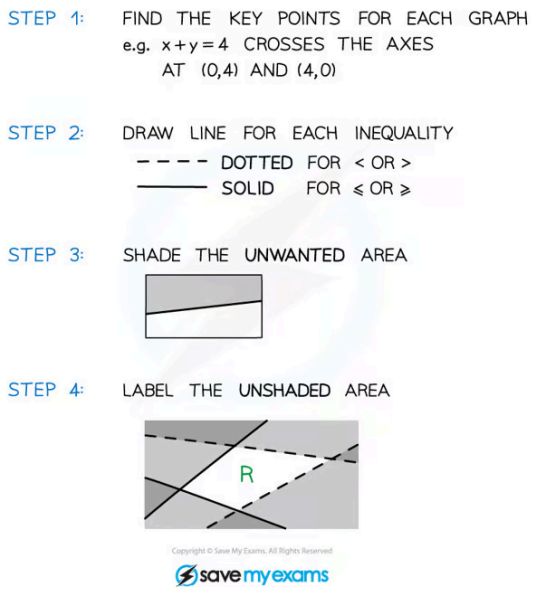
- Inequalities can be represented on graphs by shaded regions and dotted or solid lines



- These inequalities have two variables, **x** and **y**
- Several inequalities are used at once
- The solution is an **area** on a graph (often called a **region**)
- The inequalities can be **linear** or **quadratic**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Sketch each graph
  - If the inequality is **strict (< or >)** then use a **dotted line**
  - If the inequality is **weak (≤ or ≥)** then use a **solid line**
- Decide **which side** of the line **satisfies** the inequality
  - Choose a coordinate on each side and **test** it in the inequality
    - The origin is an easy point to use
  - If it satisfies the inequality then that whole side of the line satisfies the inequality
    - For example: (0,0) satisfies the inequality $y < x^2 + 1$ so you want the side of the curve that contains the origin



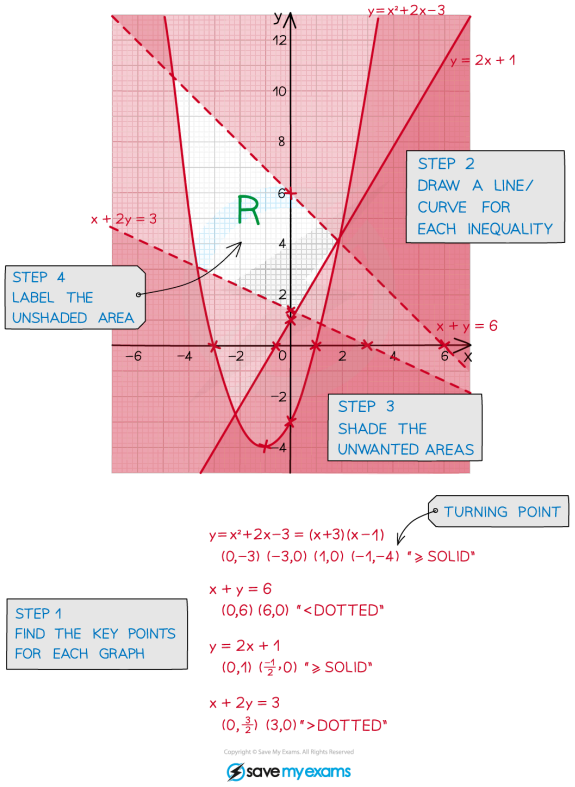
Examiner Tips and Tricks

- **Recognise** this type of inequality by the use of **two** variables
- You may have to **deduce** the inequalities from a **given** graph
- Pay careful attention to which **region** you are **asked to shade**
- Sometimes the exam could ask you to shade the region that **satisfies the inequalities, this means** you should **shade** the region that is **wanted**
- As long as your final answer is clear you should get the marks!

Worked Example

following inequalities:

$$y \geq x^2 + 2x - 3 \qquad x + y < 6$$
$$y \geq 2x + 1 \qquad x + 2y > 3$$



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